



兰州工业学院

实 验 报 告

(2023—2024 学年第 2 学期)

课程名称 《计算机网络应用》

专业班级 专升本电信 23

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实验项目	实验六 服务器的配置		
实验时间	2024 年 6 月 14 日	实验地点	信息与通信工程基础实验室 (EDA 技术实验室)

一、实验目的

1. 了解动态主机配置协议的技术背景；
2. 了解动态主机配置协议的作用；
3. 掌握 DHCP 的配置。

二、实验环境

Cisco Packet Tracer 模拟软件

三、实验步骤

1、 按如图 1 所示搭建实验环境，将 R1 配置成 DHCP 服务器，使得 PC 机能够自动获取到 200.192.1.0/24 网段的 IP 地址。其中，200.192.1.3~200.192.1.13 不可用。另外，地址池名称为学生姓名（张三的为 zhangsan）。

本实验要求通过对 DHCP 动态主机配置，实现拓扑中全网的连通性。

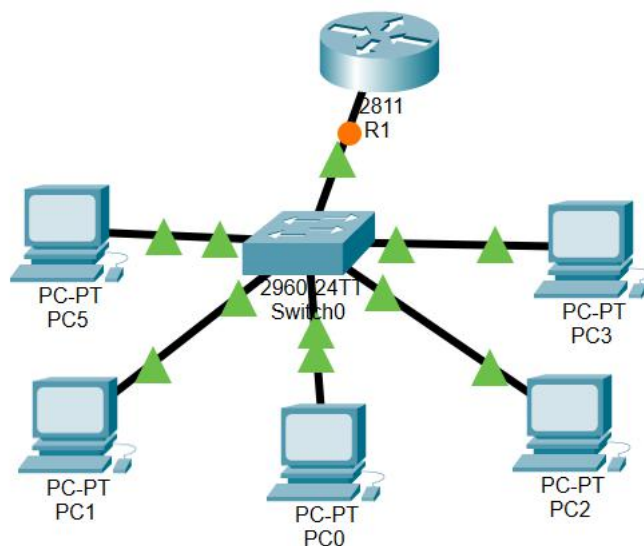


图 1 实验拓扑结构图

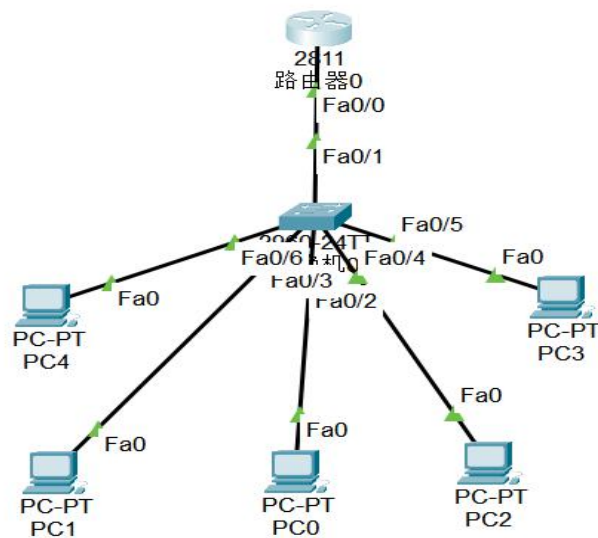


图 1 在 Cisco Packet Tracer 模拟软件的拓扑图

端口号: fa0/0
地址: 202.192.1.1

图 2 路由器连接的端口信息（端口号及相应地址）

```
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip add
Router(config-if)#ip address 202.192.1.1 255.255.255.0
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed
state to up

Router(config-if)#exit
Router(config)#
Router(config)#ip dh
Router(config)#ip dhcp poo
Router(config)#ip dhcp pool jiangyong
Router(dhcp-config)#net
Router(dhcp-config)#network 202.192.1.0 255.255.255.0
Router(dhcp-config)#dns
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#defa
Router(dhcp-config)#default-router 202.192.1.1
Router(dhcp-config)#exit
Router(config)#ip dh
Router(config)#ip dhcp exc
Router(config)#ip dhcp excluded-address 202.192.1.3 202.192.1.13
Router(config)#
```

图 3 DHCP 动态主机配置命令

```

C:\>ipconfig /all

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Physical Address.....: 0001.4236.510B
    Link-local IPv6 Address.....: FE80::201:42FF:FE36:510B
    IPv6 Address.....: ::
    IPv4 Address.....: 202.192.1.14
    Subnet Mask.....: 255.255.255.0
    Default Gateway.....: ::
                                202.192.1.1
    DHCP Servers.....: 202.192.1.1
    DHCPv6 IAID.....:
    DHCPv6 Client DUID.....: 00-01-00-01-47-D5-18-A8-00-01-42-36-51-0B
    DNS Servers.....: ::
                                8.8.8.8

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Physical Address.....: 000C.CF75.686C
    Link-local IPv6 Address.....: ::
--More--

```

图 4 PC0 的 IP 地址、子网掩码、网关信息、DNS

```

C:\>ipconfig /all

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Physical Address.....: 0004.9A76.AB36
    Link-local IPv6 Address.....: FE80::204:9AFF:FE76:AB36
    IPv6 Address.....: ::
    IPv4 Address.....: 202.192.1.2
    Subnet Mask.....: 255.255.255.0
    Default Gateway.....: ::
                                202.192.1.1
    DHCP Servers.....: 202.192.1.1
    DHCPv6 IAID.....:
    DHCPv6 Client DUID.....: 00-01-00-01-DD-21-20-1C-00-04-9A-76-
AB-36
    DNS Servers.....: ::
                                8.8.8.8

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Physical Address.....: 0006.2A65.8816
    Link-local IPv6 Address.....: ::
--More--

```

图 5 PC1 的 IP 地址、子网掩码、网关信息、DNS

```

C:\>ipconfig /all

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Physical Address.....: 00E0.F9C8.9AD2
    Link-local IPv6 Address.....: FE80::2E0:F9FF:FEC8:9AD2
    IPv6 Address.....: ::
    IPv4 Address.....: 202.192.1.15
    Subnet Mask.....: 255.255.255.0
    Default Gateway.....: ::
                                202.192.1.1
    DHCP Servers.....: 202.192.1.1
    DHCPv6 IAID.....:
    DHCPv6 Client DUID.....: 00-01-00-01-70-A0-7D-AA-00-E0-F9-C8-9A-
D2
    DNS Servers.....: ::
                                8.8.8.8

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Physical Address.....: 0003.E4A1.61EC
    Link-local IPv6 Address.....: ::
--More--

```

图 6 PC2 的 IP 地址、子网掩码、网关信息、DNS

```

C:\>ipconfig /all

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...: 
    Physical Address.....: 000D.BDB7.E7A0
    Link-local IPv6 Address.....: FE80::20D:BDF7:FEB7:E7A0
    IPv6 Address.....: ::
    IPv4 Address.....: 202.192.1.16
    Subnet Mask.....: 255.255.255.0
    Default Gateway.....: ::
                                202.192.1.1
    DHCP Servers.....: 202.192.1.1
    DHCPv6 IAID.....: 
    DHCPv6 Client DUID.....: 00-01-00-01-19-15-7D-CC-00-0D-BD-B7-E7-A0
    DNS Servers.....: ::
                                8.8.8.8

Bluetooth Connection:

    Connection-specific DNS Suffix...: 
    Physical Address.....: 0001.6420.20CC
    Link-local IPv6 Address.....: ::
--More--

```

图 7 PC3 的 IP 地址、子网掩码、网关信息、DNS

```

C:\>ipconfig /all

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...: 
    Physical Address.....: 00D0.FFD6.7850
    Link-local IPv6 Address.....: FE80::2D0:FFFF:FED6:7850
    IPv6 Address.....: ::
    IPv4 Address.....: 202.192.1.17
    Subnet Mask.....: 255.255.255.0
    Default Gateway.....: ::
                                202.192.1.1
    DHCP Servers.....: 202.192.1.1
    DHCPv6 IAID.....: 
    DHCPv6 Client DUID.....: 00-01-00-01-2D-E4-D6-E0-00-D0-FF-D6-78-50
    DNS Servers.....: ::
                                8.8.8.8

Bluetooth Connection:

    Connection-specific DNS Suffix...: 
    Physical Address.....: 00E0.A34A.1269
    Link-local IPv6 Address.....: ::
--More--

```

图 8 PC4 的 IP 地址、子网掩码、网关信息、DNS

```

C:\>ping 202.192.1.2
Pinging 202.192.1.2 with 32 bytes of data:
Reply from 202.192.1.2: bytes=32 time<1ms TTL=128
Reply from 202.192.1.2: bytes=32 time<1ms TTL=128
Reply from 202.192.1.2: bytes=32 time<1ms TTL=128
Reply from 202.192.1.2: bytes=32 time<1ms TTL=128
Ping statistics for 202.192.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 202.192.1.15
Pinging 202.192.1.15 with 32 bytes of data:
Reply from 202.192.1.15: bytes=32 time<1ms TTL=128
Reply from 202.192.1.15: bytes=32 time<1ms TTL=128
Reply from 202.192.1.15: bytes=32 time<1ms TTL=128
Reply from 202.192.1.15: bytes=32 time<1ms TTL=128
Ping statistics for 202.192.1.15:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 202.192.1.16
Pinging 202.192.1.16 with 32 bytes of data:
Reply from 202.192.1.16: bytes=32 time<1ms TTL=128
Reply from 202.192.1.16: bytes=32 time<1ms TTL=128
Reply from 202.192.1.16: bytes=32 time<1ms TTL=128
Reply from 202.192.1.16: bytes=32 time<1ms TTL=128
Ping statistics for 202.192.1.16:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 1ms
C:\>ping 202.192.1.17
Pinging 202.192.1.17 with 32 bytes of data:
Reply from 202.192.1.17: bytes=32 time<1ms TTL=128
Reply from 202.192.1.17: bytes=32 time<1ms TTL=128
Reply from 202.192.1.17: bytes=32 time<1ms TTL=128
Reply from 202.192.1.17: bytes=32 time<1ms TTL=128
Ping statistics for 202.192.1.17:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 2ms
C:\>

```

图 9 PC0 和其它 PC 的连通性


```

C:\>ping 202.192.1.14
Pinging 202.192.1.14 with 32 bytes of data:
Reply from 202.192.1.14: bytes=32 time<1ms TTL=128
Reply from 202.192.1.14: bytes=32 time<1ms TTL=128
Reply from 202.192.1.14: bytes=32 time<1ms TTL=128
Reply from 202.192.1.14: bytes=32 time<1ms TTL=128
Ping statistics for 202.192.1.14:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 202.192.1.15
Pinging 202.192.1.15 with 32 bytes of data:
Reply from 202.192.1.15: bytes=32 time<1ms TTL=128
Reply from 202.192.1.15: bytes=32 time<1ms TTL=128
Reply from 202.192.1.15: bytes=32 time<1ms TTL=128
Reply from 202.192.1.15: bytes=32 time<1ms TTL=128
Ping statistics for 202.192.1.15:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 202.192.1.16
Pinging 202.192.1.16 with 32 bytes of data:
Reply from 202.192.1.16: bytes=32 time<1ms TTL=128
Reply from 202.192.1.16: bytes=32 time<1ms TTL=128
Reply from 202.192.1.16: bytes=32 time<1ms TTL=128
Reply from 202.192.1.16: bytes=32 time<1ms TTL=128
Ping statistics for 202.192.1.16:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 202.192.1.17
Pinging 202.192.1.17 with 32 bytes of data:
Reply from 202.192.1.17: bytes=32 time<1ms TTL=128
Reply from 202.192.1.17: bytes=32 time<1ms TTL=128
Reply from 202.192.1.17: bytes=32 time<1ms TTL=128
Reply from 202.192.1.17: bytes=32 time<1ms TTL=128
Ping statistics for 202.192.1.17:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 8ms, Average = 2ms

```

图 10 PC1 和其它 PC 的连通性

```

C:\>ping 202.192.1.14
Pinging 202.192.1.14 with 32 bytes of data:
Reply from 202.192.1.14: bytes=32 time<1ms TTL=128
Reply from 202.192.1.14: bytes=32 time<1ms TTL=128
Reply from 202.192.1.14: bytes=32 time<1ms TTL=128
Reply from 202.192.1.14: bytes=32 time<1ms TTL=128
Ping statistics for 202.192.1.14:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 202.192.1.2
Pinging 202.192.1.2 with 32 bytes of data:
Reply from 202.192.1.2: bytes=32 time=10ms TTL=128
Reply from 202.192.1.2: bytes=32 time<1ms TTL=128
Reply from 202.192.1.2: bytes=32 time<1ms TTL=128
Reply from 202.192.1.2: bytes=32 time<1ms TTL=128
Ping statistics for 202.192.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms
C:\>ping 202.192.1.16
Pinging 202.192.1.16 with 32 bytes of data:
Reply from 202.192.1.16: bytes=32 time<1ms TTL=128
Reply from 202.192.1.16: bytes=32 time<1ms TTL=128
Reply from 202.192.1.16: bytes=32 time<1ms TTL=128
Reply from 202.192.1.16: bytes=32 time<1ms TTL=128
Ping statistics for 202.192.1.16:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 202.192.1.17
Pinging 202.192.1.17 with 32 bytes of data:
Reply from 202.192.1.17: bytes=32 time<1ms TTL=128
Reply from 202.192.1.17: bytes=32 time=6ms TTL=128
Reply from 202.192.1.17: bytes=32 time<1ms TTL=128
Reply from 202.192.1.17: bytes=32 time<1ms TTL=128
Ping statistics for 202.192.1.17:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 6ms, Average = 1ms

```

图 11 PC2 和其它 PC 的连通性

```

C:\>ping 202.192.1.14
Pinging 202.192.1.14 with 32 bytes of data:
Reply from 202.192.1.14: bytes=32 time<1ms TTL=128
Reply from 202.192.1.14: bytes=32 time<1ms TTL=128
Reply from 202.192.1.14: bytes=32 time<1ms TTL=128
Reply from 202.192.1.14: bytes=32 time<1ms TTL=128
Ping statistics for 202.192.1.14:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 202.192.1.2
Pinging 202.192.1.2 with 32 bytes of data:
Reply from 202.192.1.2: bytes=32 time<1ms TTL=128
Reply from 202.192.1.2: bytes=32 time<1ms TTL=128
Reply from 202.192.1.2: bytes=32 time<1ms TTL=128
Reply from 202.192.1.2: bytes=32 time<1ms TTL=128
Ping statistics for 202.192.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 202.192.1.15
Pinging 202.192.1.15 with 32 bytes of data:
Reply from 202.192.1.15: bytes=32 time<1ms TTL=128
Reply from 202.192.1.15: bytes=32 time<1ms TTL=128
Reply from 202.192.1.15: bytes=32 time<1ms TTL=128
Reply from 202.192.1.15: bytes=32 time<1ms TTL=128
Ping statistics for 202.192.1.15:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 202.192.1.17
Pinging 202.192.1.17 with 32 bytes of data:
Reply from 202.192.1.17: bytes=32 time<1ms TTL=128
Reply from 202.192.1.17: bytes=32 time<1ms TTL=128
Reply from 202.192.1.17: bytes=32 time<1ms TTL=128
Reply from 202.192.1.17: bytes=32 time<1ms TTL=128
Ping statistics for 202.192.1.17:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

```

图 12 PC3 和其它 PC 的连通性

```

C:\>ping 202.192.1.14
Pinging 202.192.1.14 with 32 bytes of data:
Reply from 202.192.1.14: bytes=32 time<1ms TTL=128
Reply from 202.192.1.14: bytes=32 time<1ms TTL=128
Reply from 202.192.1.14: bytes=32 time<1ms TTL=128
Reply from 202.192.1.14: bytes=32 time=6ms TTL=128
Ping statistics for 202.192.1.14:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 6ms, Average = 1ms
C:\>ping 202.192.1.2
Pinging 202.192.1.2 with 32 bytes of data:
Reply from 202.192.1.2: bytes=32 time<1ms TTL=128
Reply from 202.192.1.2: bytes=32 time=5ms TTL=128
Reply from 202.192.1.2: bytes=32 time<1ms TTL=128
Reply from 202.192.1.2: bytes=32 time<1ms TTL=128
Ping statistics for 202.192.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 5ms, Average = 1ms
C:\>ping 202.192.1.15
Pinging 202.192.1.15 with 32 bytes of data:
Reply from 202.192.1.15: bytes=32 time<1ms TTL=128
Reply from 202.192.1.15: bytes=32 time<1ms TTL=128
Reply from 202.192.1.15: bytes=32 time<1ms TTL=128
Reply from 202.192.1.15: bytes=32 time<1ms TTL=128
Ping statistics for 202.192.1.15:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 202.192.1.16
Pinging 202.192.1.16 with 32 bytes of data:
Reply from 202.192.1.16: bytes=32 time<1ms TTL=128
Reply from 202.192.1.16: bytes=32 time<1ms TTL=128
Reply from 202.192.1.16: bytes=32 time<1ms TTL=128
Reply from 202.192.1.16: bytes=32 time<1ms TTL=128
Ping statistics for 202.192.1.16:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>

```

图 13 PC4 和其它 PC 的连通性

四、总结

实验成绩评定表

序号	考核项目	分值分布	成绩
1	出勤与纪律情况	10	
2	实验任务完成情况	40	
3	实验报告质量	50	
总分			
指导教师签字			

备注：考核项目及分值分布应依据教学大纲确定。